Game Theoretic Merging Behavior Control for Autonomous Vehicle at Highway On-Ramp

Chao Wei, Yuanhao He, Hanqing Tian, and Yanzhi Lv

***Abstract*— In the most of previous studies, social interactions between vehicles are not considered explicitly when designing decision making and motion planning module. Nevertheless, the strong interaction exists in the most of driving scenarios, especially in highway junction. This paper presents a game theoretic merging behavior control system for autonomous vehicle at on-ramp junction considering the interaction between the merging vehicle and following vehicle in the main lane. In this work, a driving style estimation is proposed to deal with the heterogeneity of driving style. Then we adopt a model predictive control (MPC) method to plan the optimal merging trajectory based on the game theoretic decision making result. Finally, simulation and Human-in-the-loop (HIL) experiment result shows the effectiveness of our approach in on-ramp merging scenario.**

***Index Terms*— Merging behavior, game theory, autonomous vehicle, model predictive control, driving style.**

I. Introduction

AUTONOMOUS driving has become a core technology for improving driving safety and road traffic. Some key techniques such as behavior prediction and motion planning still remain challenging, especially in urban scenario or at highway junction. In on-ramp area, traditional decision making and motion planning system does not consider interactions between autonomous vehicles explicitly, which differs from the way that human drivers interact with environment [1], thus does not make full use of information in driving environment. Game theory is an effective approach to deal with vehicle interactions in traffic [2]. However, ensuring the merging decision model accuracy with the consideration of behaviors of surrounding vehicles is one main challenge of applying game theory approach to autonomous vehicle. In terms of motion planning, optimization-based method allows continuous planning thus generates better solution than discretized method.

In this paper, we develop a merging behavior control (MBC) system utilizing game theory and model predictive control. The contributions of this paper are summarized as follow:

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1. This paper proposes a merging decision making model for autonomous vehicle considering the interaction between autonomous vehicle and following vehicle based on game theory. In addition, a real-time and continuous driving style estimation algorithm is proposed to consider driver's heterogeneity, which is significant for interactive decision making.

2. The MBC system tightly combines the game theoretic decision making module and Model Predictive Control (MPC) motion planner to generate a feasible merging trajectory in real time.

3. Simulation and human-in-the-loop(HIL) experiment are conducted to validate the precision of driving style estimation algorithm and the ability of the MBC system when interacting with human.

The paper is organized as follow: Section II reviews the merging decision modelling and motion planning approaches. The problem is formulated in Section III. Decision making and motion planning approach are described in Section IV and Section V, respectively. In Section VI, the HIL simulation and result analysis are given. Finally, the conclusions are drawn in Section VII.

II. Related Works

A. On-Ramp Merging Decision Modeling

On-ramp merging counts as one of the most challenging tasks for autonomous vehicles [3]. Not only merging vehicle's behavior but also surrounding vehicles' behaviors affects the merging process [4]. On-ramp merging maneuver is a typical case of mandatory lane change(MLC) which requires autonomous vehicle to merge before the acceleration lane ends on account of lane constraint. Major MLC decision model are divided into 4 categories: rule-based model [5], utility theory model [6], Markov process model [7]. The drivers' behavior in the surrounding vehicles differs when the lane changing vehicle accept or reject lane changing. For instance, in a merging scenario, the merging vehicle choose either to wait or to merge, while the following vehicle on the main lane responds to merging vehicle by yielding (by decelerating) or by discouraging the lane changing (by accelerating). This shows a strong interaction between both participant. Each participant's decision depends on the respond of other, therefore accident may occur without considering decision of either participant. The above mentioned types of model does not explicitly consider the interaction, which may ignore some potential information, thus the decision may not be optimal.

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Game theory performs effectively to deal with the interaction problem due to the explicit consideration of the interaction. Game theory-based approach incorporates both decisions of the lane-changing vehicle and following vehicle in the target lane explicitly. The decisions of each game participants depend on the action of other. In [8], Kita proposed the first game theory-based MLC model. The study considers merging-giveaway interaction between merging vehicle and following vehicle as a non-cooperative non-zero-sum game. In this model, time-to-collision (TTC) is taken as the safety criterion, which shows promise in modeling traffic flow. Talebpour *et al.* proposed a game theoretic model for connected environment [9]. They applied a Harsanyi transformation [10] to transform the incomplete information game to imperfect information game. Wang *et al.* [11] developed a differential game based controller to control a given vehicle's car-following and lane-changing behavior. This study does not assume a set of finite strategy, but calculate different combination of acceleration and lane changing time to minimize the cost function. In Kang and Rakha's merging model, merging vehicle has three strategies: merging, waiting and overtaking, and strategies for following vehicle are yielding or not [12]. The payoff of each player are established based on safety, traveling efficiency and acceleration. The most common type used in MLC modeling is static game, and a Nash equilibrium can be found. The above mentioned approaches consider a pure Nash equilibrium, and in other works [13]–[14], the mixed Nash solutions are considered. In recent year, advanced game types such as Stackelberg game, differential game and dynamic game are adopted. Yu *et al.* developed a human-like lane changing model based on Stackelberg game, and aggressiveness of competing driver is estimated and updated online as a parameter in the payoff function of the lane changing vehicle [15]. In [1], a Stackelberg game is tightly combined with MPC, so the game and MPC can cooperate closely with each other. Arbis and Dixit introduced Quantal Response Equilibrium (QRE) to model merging [16]. In QRE, drivers' anticipations are probabilistic due to the errors in perception which is unavoidable in real driving behavior. In [17], a dynamic game approach is employed to determine an optimal control sequence. To simplify the problem, interaction-unaware maneuver probabilities are used to approximate the maneuver behavior. In [18], Fukuyama utilized a zero-suppressed binary decision diagram to solve the connected autonomous vehicle (CAV) merging control problem.

The above mentioned on-ramp merging decision modeling can be categorized into 2 types by their strategy. One type of the decision modeling only considers the rough maneuver such as wait, merge, accelerating, and decelerating. Another type considers the physical quantity e.g. acceleration, waiting time, and speed profile. For merging behavior controller, giving only rough instructions does not guarantee the safe and efficient motion planning.

B. Motion Planning

Motion planning approaches can be divided into four categories: search-based [19], [20], sampling-based [21], [22], and optimization-based method. Search-based method is capable of obstacle avoidance, however, the search space significantly affects the computational cost on account of "curse of dimensionality". Sampling-based method such as Rapidly-exploring random trees (RRT) [21] simplifies the larger problem by sampling patterns. Optimization-based and learning-based method can overcome above deficiency. Optimization-based method formulates the planning problem as a numerical optimization problem. It is capable of planning in continuous space, and the multiple objective can be evaluated [23]. Typical optimization-based methods include direct collocation and indirect shooting [24]. Direct collocation method considers trajectory points as decision variables, and optimize trajectory by directly changing trajectory points. Sequential Quadratic programming (SQP) is a common tool for this method [25], [26]. Another type of optimization-based method is the shooting method. This method considers control input as decision variables, which is related to optimal control. Differential Dynamic Programming (DDP) has good performance in solving optimal control problem [27], of which Iterative Linear Quadratic Regulator (ILQR) is a fast version [28]. In [23], a constrained ILQR is proposed to deal with general form of constraint.

Many research works have estimated merging problem. Vehicle-to-vehicle (V2V) communication technology forms a transportation system, in which connected autonomous vehicles(CAVs) are controlled to pass through merging area. In [29]–[32], trajectory planning methods for CAVs in on-ramp zone are presented on the premise that CAVs can share information with each other through V2V technology. When V2V communication is unavailable, an online approach can be useful to deal with imperfect observation. In [2], the study developed an MPC trajectory planning method with the consideration of multiple decision such as time to collide (TTC), acceleration, etc. Masakazu Mukai *et al.* present a receding horizon style motion planning method for on-ramp merging incorporating mixed integer programming (MIP) [33]. In [34], the study exploits main lane driver's acceptance for the merging vehicle into an MPC framework, thus the proposed approach can generate reasonable behavior in merging zone. In [35], [36], Hang *et al.* take social interaction into consideration, and propose an integrated framework for lane changing control.

C. Limitation in Previous Studies

The above literature review of on-ramp merging decision making and motion planning reveals some limitation in the previous studies.

First, some of previous works propose improper assumptions on driving style estimation. In [36], the authors assume that the autonomous vehicle well knows driving style of other vehicles. Indeed, autonomous vehicle can obtain the information of surrounding vehicles through V2V communication. Nevertheless, V2V communication will not be available on all vehicles in the near future, thus it is more appropriate to assume that autonomous vehicle estimates surrounding vehicle's driving style by playing game with it through the

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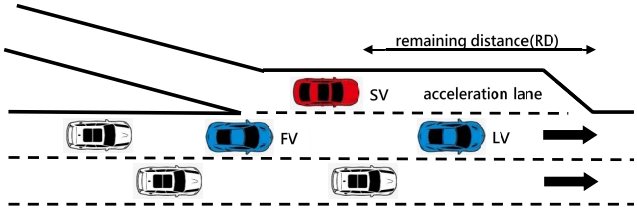

Fig. 1. Schematic representation of merging scenario.

merging maneuver. Also, some works divide driving styles into discrete categories which is not accurate, as the driving style still different among same type of drivers.

Second, few studies implement the game theoretic algorithm in HIL simulation. As the decisions vary among different drivers, guaranteeing the effectiveness of the game theoretic algorithm remains a challenge. The HIL simulation test verifies the performance of game theoretic algorithm in both decision making and decision prediction. The experiment designed in some of previous studies do not include human's participation, which could not validate the performance of game theoretic method. Moreover, HIL simulation test also validates the computational efficiency of the algorithm. Thus the HIL simulation is an indispensable for studying interactive decision making and motion planning.

In summary, most of the above decision modeling approaches do not consider the corresponding decision of following vehicle explicitly. In merging scenario, more than one vehicle participates in the merging maneuver, the decision controller implicitly considering the interaction does not make full use of information from surrounding vehicles. Furthermore, improper assumption and incomplete experiment verification exists in most of game theoretic approach. Therefore, it is urgent to develop a more reasonable decision making and motion planning system for ramp merging scenario.

III. Problem Formulation

The merging scenario is shown in Fig 1, a subject vehicle (SV) in the acceleration lane; a currently following vehicle (FV) in the main lane; and a lead vehicle (LV) in the main lane. SV is an autonomous vehicle which needs to execute an MLC operation before the acceleration lane ends. The goal of this study is to find the optimal merging solution for SV during the merging process.

In the previous works, decision making and motion planning are developed separately, which would not find the optimal solution easily [35]. To solve this problem, we utilize decision making and motion planning in an integrated way. First, we design a game theoretic decision making module incorporating the interaction between SV and FV. Decision making module yields the optimal decision of FV (i.e. the optimal acceleration and the optimal waiting time before merging) and the predicted decision of FV (i.e., the acceleration). Based on the decision result, the motion planning module plans a feasible trajectory in receding horizon manner. Fig. 2 illustrates the proposed integrated control diagram.

Driving style recognition approaches includes rule-based, model-based and learning-based approach [37]. Most studies on driving style estimation utilize the field data, which take

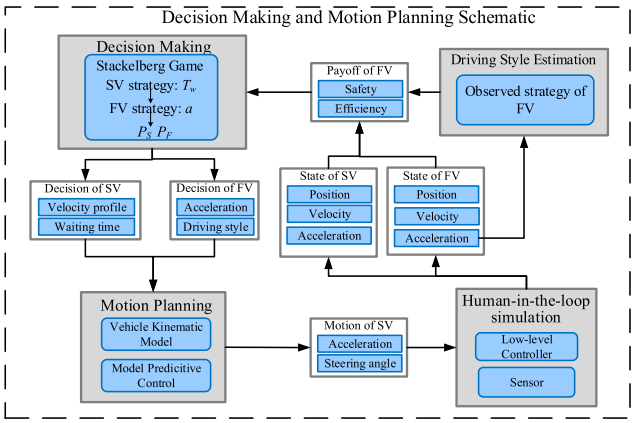

Fig. 2. Merging behavior control framework.

both human factor (e.g. age, behavior and aggression potential) and environment factor (e.g. traffic, road condition and sight distance) into consideration. Nevertheless, approaches incorporating field data are not able to be utilized for autonomous vehicle as it is difficult to observe and process the data on-line. Some approaches classify the driving style into a discrete category based on the distribution of selected parameter. For instance, driving style is often divided into three types: aggressive, moderate and calm. Discrete category could not evaluate the driving style quantitatively as the driving style of same type of drivers still differ from each other. Thus we adopt a quantitative driving style estimation algorithm that only use simple data, which can estimate driving style of driver in surrounding vehicle. This study adopts an approach similar to the algorithm proposed in [1]. In addition, we remove the unrealistic assumption that drivers have discrete and fixed driving style. Instead, this study estimate driving style based on the driver's action in real time, thus the estimated driving style can be more accurate.

In this study, we make the following assumption: (i) Both SV and FV are assumed to behave in a rational way. Under this assumption the game between SV and FV is established. (ii) This study does not consider SV's strategy of overtaking LV. Because the overtaking is dangerous, and overtaking maneuver during merging process is forbidden in many countries. (iii) FV's maneuver of changing lane is rarely observed in the field data [38], thus FV's lane changing strategy is not considered here. Based on these assumptions, decision making and motion planning module will be discussed in the following section.

IV. Game Theoretic Decision Making

Game theory provides a solution for interactive decision making. Under the assumption made in the last section, we consider a non-cooperative Stackelberg game between SV and FV for merging scenario.

A. On-Ramp Merging Decision Modeling

Stackelberg game is proposed in 1934 by the German economist Heinrich Freiherr von Stackelberg [39]. Stackelberg game is a two-player two-stage game in which the leader first

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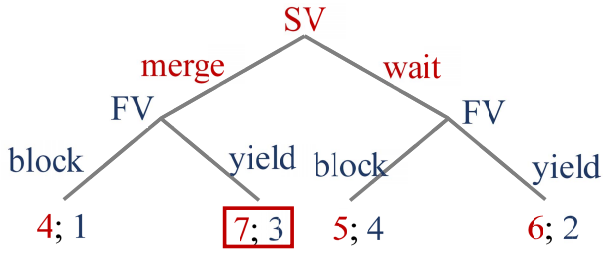

Fig. 3. Payoffs for SV and FV under Stackelberg game theory.

adopts his strategy, and the follower take strategy to respond to the leader. Each player in the game aims to maximize his own payoff, and the leader is considered to be able to predict the follower's strategy given the leaders's own strategy. Stackelberg game can be regarded as a bilevel optimization problem. The high level is the strategy adopted by the leader, and the low level is the follower's strategy under the given the leader's strategy. The solution of this bilevel optimization problem is Stackelberg Equilibrium. Similar to [1], the SE can be defined as:

$$s_L^* = \arg\max_{s_L \in S_L} \left(\min_{s_F \in S_F^*} P_L(s_L, s_F) \right) \tag{1}$$

$$S_F^*(s_L) \triangleq \{ s_F^* \in S_F : P_F(s_L, s_F^*) \geq P_F(s_L, s_F), \forall s_F \in S_F \} \tag{2}$$

where the subscript L and F denote the leader and the follower, s_L denotes the leader's strategy and s_F denotes the follower's strategy, s_L^* and s_F^* denotes leader's and follower's optimal strategy, respectively. S_L and S_F are leader's and follower's strategy spaces, S_F^* is follower's optimal strategy set given the leader's strategy, P_L and P_F are leader's and follower's payoffs.

In this study, we model the merging maneuver into the above mentioned general form of Stackelberg game. Here we take a simple example to illustrate the process of Stackelberg game in merging scenario. As is shown in Fig. 1, SV's strategies are merging or waiting, and FV's chooses to yield or block according to the strategy adopted by SV. SV and FV choose their strategies and get corresponding payoffs. The payoffs in this game are set as shown in Fig. 3. Both SV and FV try to maximize their own payoffs. SV first adopts the strategy of merging or waiting. If SV choose to merge, FV will choose to yield because the payoff of yielding is higher. And if SV choose to wait, FV will choose block due to the higher payoff. Thus SV would choose (merge, yield) as a result of evaluation for its own payoff. From this example we can find that though SV takes the priority to decide first, its own payoff also depends on FV's strategy.

Next, we will introduce three elements in game theoretic merging model: players, strategies and payoffs.

(1) Players: In merging scenario, we define SV and FV as the leader and follower, respectively. SV first chooses its strategy by turning on the turn signal when travels on the acceleration lane. Then FV would adopt an acceleration to respond to SV.

(2) Strategies: SV's strategies are waiting time before merging T_w; For FV, the strategy is its acceleration a_{FV}.

(3) Payoffs: SV's payoff consists of three parts: longitudinal payoff, lateral payoff and acceleration payoff. FV's payoff includes safety payoff, travel efficiency payoff and acceleration payoff.

The detail of SV and FV will be introduced in Section IV-B and Section IV-C.

B. Payoff for Subject Vehicle

In this study, we assume that SV wait in the acceleration lane from step 1 to N_w. N_w is defined as: $N_w = T_w/\tau$. Where T_w is SV's waiting time before merging, and τ is time resolution.

SV's longitudinal speed profile candidates can be generated by cubic polynomial speed curve. Then the jerk, acceleration, and longitudinal displacement can be calculated.

SV's payoff includes longitudinal part, lateral part, acceleration part and jerk part, which is given as follow:

$$P_{SV} = \sum_{k=1}^{N} [\sigma k_{lon}^{SV} P_{lon}^{SV}(k) + (1-\sigma) k_{lat}^{SV} P_{lat}^{SV}(k) + k_{acc}^{SV} P_{acc}^{SV}(k) + k_{jerk}^{SV} P_{jerk}^{SV}(k)] \tag{3}$$

where

$$\sigma = \begin{cases} 1, & k \leq N_w \\ 0, & k > N_w \end{cases} \tag{4}$$

P_{lon}^{SV}, P_{lat}^{SV}, P_{acc}^{SV} and P_{jerk}^{SV} are SV's longitudinal payoff, lateral payoff, acceleration payoff and jerk payoff, respectively. k_{lon}^{SV}, k_{lat}^{SV}, k_{acc}^{SV} and k_{jerk}^{SV} are corresponding weighting coefficient. N denotes the total steps in the predicting horizon. And σ denotes SV's state, at the step before merging, σ equals 1, otherwise σ equals 0.

P_{lon}^{SV} and P_{lat}^{SV} are two main part of SV's payoff. We design P_{lon}^{SV} and P_{lat}^{SV} as piecewise functions in order to measure them in the same metric. Each part of SV's payoff is expressed as follow:

$$P_{lon}^{SV}(k) = \begin{cases} RD(k)/RD_{thr}(k) - 2, & v_{SV}(k)t_r \geq RD(k) \\ -2, & v_{SV}(k)t_r < RD(k) \end{cases} \tag{5}$$

$$P_{lat}^{SV}(k) = \begin{cases} 1, & \Delta x(k) > \Delta x_{thr} \text{ or } \Delta x(k) < -\Delta x_{thr} \\ -\Delta x(k)/\Delta x_{thr}, & \Delta x(k) \geq -\Delta x_{thr} \text{ and } \Delta x(k) < -\Delta x_{sf} \\ \Delta x(k)/\Delta x_{thr}, & \Delta x(k) \geq \Delta x_{sf} \text{ and } \Delta x(k) < \Delta x_{thr} \\ -1, & otherwise \end{cases} \tag{6}$$

$$P_{acc}^{SV}(k) = -a_{SV}(k)^2, \quad P_{jerk}^{SV}(k) = -j_{SV}(k)^2 \tag{7}$$

$$\Delta x(k) = x_{SV}(k) - x_{FV}(k) \tag{8}$$

In equation (5) -(8), v_{SV} is SV's velocity, t_r is the SV's react time (i.e. 2s), RD is the remaining distance on the acceleration lane, and RD is a constant that determines the threshold of longitudinal payoff function. Δx is the longitudinal distance between SV and FV. Δx_{thr} and Δx_{sf} are the threshold distance and safety distance. A_{SV} is the longitudinal acceleration of SV,

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and j_{SV} is the longitudinal jerk of SV. $j_{SV}(k) = [a_{SV}(k) - a_{SV}(k-1)]/\tau$. Note that a_{SV} here is a variable rather than the observed acceleration from sensors.

SV's longitudinal payoff function indicates that SV intends to merge as soon as possible. And lateral payoff indicates that SV aims to maintain a safe distance relative to FV. P_{acc} and P_{jerk} are the costs of driving comfort. SV's payoff reveals that SV has to strike the balance between quick merging maneuver and driving safety. Note that SV is not able to observe SV's strategy T_w, here we use the decreasing acceleration model for FV to determine the time when SV will get a safe position to change lane, which is the same as the approach in [1].

C. Payoffs for Following Vehicle

Similar to SV's payoff, FV's payoff consists of three parts: driving safety payoff, travel efficiency payoff and acceleration payoff:

$$P_{FV} = \sum_{k=1}^{N} [\rho k_{sf}^{SV} P_{sf}^{SV}(k) + (1-\rho) k_{eff}^{SV} P_{eff}^{SV}(k) + k_{acc}^{SV} P_{acc}^{SV}(k)] \quad (9)$$

P_{sf}^{FV}, P_{eff}^{FV} and P_{acc}^{FV} are driving safety payoff, travel efficiency payoff and acceleration payoff,, respectively. k_{sf}^{FV}, k_{eff}^{FV} and k_{acc}^{FV} are corresponding weighting coefficient. ρ is a parameter between 0 and 1, which indicates driver's driving style in FV. The P_{sf}^{FV}, P_{eff}^{FV} and P_{acc}^{FV} are given as:

$$P_{sf}^{FV}(k) = \begin{cases} |\Delta x(k)| / \Delta x_{thr}, & |\Delta x(k)| < \Delta x_{thr} \\ 1, & |\Delta x(k)| \geq \Delta x_{thr} \end{cases} \quad (10)$$

$$P_{eff}^{FV}(k) = \begin{cases} 1, & \Delta x(k) < 0 \\ -\Delta x(k)/\Delta x_{thr} + 1, & \Delta x(k) < \Delta x_{thr} \\ 0, & \Delta x(k) \geq \Delta x_{thr} \end{cases} \quad (11)$$

$$P_{acc}^{FV}(k) = -a_{FV}(k)^2 \quad (12)$$

$$\Delta x(k) = x_{SV}(k) - x_{FV}(k) \quad (13)$$

In equation (10) -(13), Δx is the longitudinal distance between SV and FV, Δx_{thr} is the threshold distance that determines piecewise functions in (10) and (11). a_{FV} is the longitudinal acceleration of FV. Similarly, FV's acceleration is also a variable instead of the measured value.

FV's safety payoff function in (10) shows that FV's intention of keeping a relative long distance from SV. Within the threshold distance, longer distance brings higher safety payoff. FV's travel efficiency payoffs indicates that FV need to overtake SV in order to get the high travel efficiency payoff. ρ is a parameter between 0 and 1 that estimates FV's driving style. Aggressive driver in FV prefers higher driving efficiency, while cautious drivers choose to keep relative far from SV. The more aggressive the driver in FV is, the smaller the estimated parameter ρ is.

It is worth noting that driver's style in FV is not available to SV, SV need to estimates FV's driving style by playing game with it.

D. Driving Style Estimation

In some previous works, driving style of surrounding vehicle is assumed to be available, which is not realistic. It is more appropriate to assume that autonomous vehicle estimate driving style of surrounding vehicle by observing their motion online. In this paper, we adopt a driving style estimation approach similar to the algorithm proposed in [14].

Here we assume that SV is able to observe the accurate acceleration of FV. In such a way SV can observe FV's intention and make decision based on FV's motion.

The predicted acceleration of FV is a discrete array, but the SV's real acceleration is continuous, which may not be included in the strategy array. SV estimates FV's driving style by assuming that FV get a higher or equal payoff by adopting the real strategy than adopting the predicted strategy by SV. FV's driving style can be estimated by solving the following inequality:

$$P_{FV}(\rho, s_{SV}, s_{FVobserved}) \geq P_{FV}(\rho, s_{SV}, s_{FVpredicted}) \quad (14)$$

where s_{SV} is SV's strategy, $s_{FVobserved}$ is the observed strategy of FV, and $s_{FVobserved}$ is the predicted strategy of FV calculated by SV. Initially, SV does not know FV's driving style, thus FV's driving style is considered as moderate ($\rho = 0.5$). SV and FV choose their own strategy, then SV is able to observe FV's strategy and estimate FV's driving style by solving (14). If the difference is larger than a threshold, SV updates the driving style of FV.

E. Decision Making Considering Interaction

Here, we replace the abstract variables in (1) and (2) with meaningful variables. Finally, the bilevel optimization problem of merging decision making can be expressed as follow:

$$T_w^* = \arg\max_{T_w} \left(\min_{a_{FV} \in \Upsilon(T_w)} P_{SV}\left(v_{SV}^*, T_w, a_{FV}, \rho\right) \right) \quad (15)$$

$$\Upsilon(T_w) \triangleq \{a_{FV}^* \in \Omega : P_{FV}\left(v_{SV}, T_w, a_{FV}^*, \hat{\rho}\right) \geq P_{FV}\left(v_{SV}, T_w, a_{FV}, \hat{\rho}\right), \forall a_{FV} \in \Omega\} \quad (16)$$

where v_{SV} denotes SV's possible speed profile, T_w is SV's waiting time for merging, v_{SV}^* is SV's optimal speed profile, and T_w^* is the optimal waiting time. $\Upsilon(a_{SV}, T_w)$ denotes FV's optimal strategy candidates given SV's strategy, Ω denotes FV's strategy candidates. ρ is SV's driving style, and $\hat{\rho}$ indicates the estimated driving style. As the estimation of FV's driving style may not be perfect, FV should keep a safe distance from FV when FV chooses to merge. In other word, when FV's strategy T_w^* equals to 0, if distance between SV and FV, SV choose to merge, otherwise SV waits by adjusting its velocity.

In this work, we adopt discrete strategies and use exhaustive search to solve the above bilevel optimization. Here, SV's and FV's strategy are defined as follow:

$$T_w = \{0, 1, 2, 3, 4\}$$
$$a_{FV} = \{0, \pm 0.2, \pm 0.4, \ldots, \pm 2.0\}$$

In our approach, one decimal place precision is enough and under this accuracy the SE can be found in real time.

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In some scenarios where high precision is required, the bilevel optimization problem can be solved by the algorithm proposed in [40].

V. Receding Horizon Motion Planning

In this section, the motion planning algorithm for merging scenario is proposed in a Nonlinear Model Predictive Control (NMPC) style. We utilize the NMPC-based method to plan the optimal trajectory for SV's merging maneuver. In the last section, the game theoretic decision making approach generates SV's decision and predicts FV's decision. The motion planning module for SV is designed based on SV's and FV's decision from decision making module.

A. Motion Planning Model

In this work, we adopt a 2 degree-of-freedom (DOF) kinematic model for motion planning. The model is expressed as:

$$\dot{X}(t) = v(t)\cos\theta(t) \tag{17a}$$
$$\dot{Y}(t) = v(t)\cos\theta(t) \tag{17b}$$
$$\dot{\theta}(t) = v(t)\tan\delta(t)/L \tag{17c}$$
$$\dot{v}(t) = a(t) \tag{17d}$$
$$\dot{\delta}(t) = \varepsilon(t) \tag{17e}$$

where X is the longitudinal position of SV's rear axle center, Y the lateral position of SV's rear axle center, ε is the velocity of SV's rear axle center. θ is SV's heading angle, δ is SV's steering angle, L is SV's wheelbase, a is SV's acceleration, and ε is SV's steering angular velocity. a and ε are control variables.

There are some constraints that SV should not violate. 2 types of constraints are considered in this work: vehicle performance constraint and safety constraint. The two types of constraints are expressed as:

$$v_{\min} \le v(t) \le v_{\max} \tag{18a}$$
$$a_{\min} \le a(t) \le a_{\max} \tag{18b}$$
$$-\delta_{\max} \le \delta(t) \le \delta_{\max} \tag{18c}$$
$$-\varepsilon_{\max} \le \varepsilon(t) \le \varepsilon_{\max} \tag{18d}$$
$$x_{SV}(t) - X(t) \ge d_{\min} \tag{19a}$$
$$X(t) - x_{SV}(t) \le d_{\min} \tag{19b}$$

where v_{min} and v_{max} are the minimum and maximum velocity, a_{min} is the largest deceleration, and a_{max} is the largest acceleration. δ_{min} and δ_{max} are the minimum and maximum steering angle. ε_{min} and ε_{max} are the minimum and maximum steering angular velocity.

B. Motion Planning Incorporating Decision Making

The decision module generates the longitudinal displacement and waiting time before merging. To follow the desired trajectory given by decision making. The cost function is given as follow:

$$J_m = \int_0^T (k_x(x(t) - x_{tar})^2 + k_y(y(t) - y_{tar})^2 + k_a a(t)^2 + k_\delta \delta(t)^2 + k_\varepsilon \varepsilon(t)^2)dt + k_\theta \theta(T)^2 \tag{20}$$

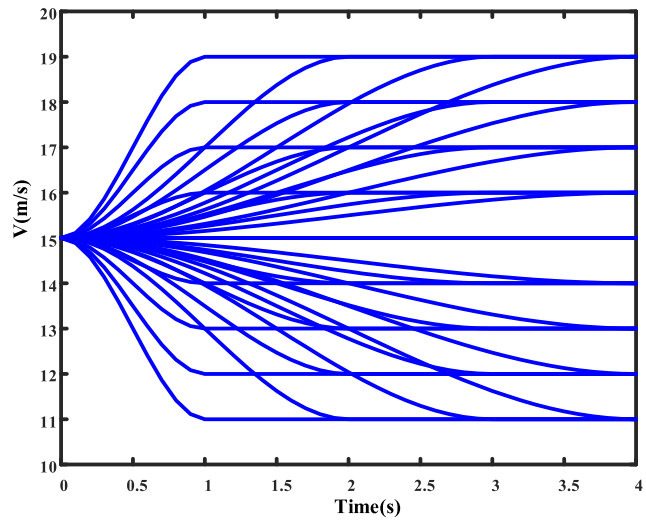

Fig. 4. Cubic polynomial longitudinal speed curve.

Fig. 5. HIL simulation environment.

where x_{tar} is the target longitudinal displacement, y_{tar} is the target lateral displacement, if waiting time is greater than 0, $y_{tar} = 0$, otherwise, $y_{tar} = y_{mainlane}$. k_x, k_y, k_a, k_δ, k_ε, k_θ are weighting coefficients.

C. Solution of Motion Planning

The motion planning model we established is a NMPC problem. To obtain the optimal control sequence, the NMPC problem is transformed into a nonlinear programming (NLP) problem by adopting single shooting method. Then we utilize sequential quadratic programming (SQP) method to solve the NLP problem. In this work, the celebrated C++ NMPC toolkit ACADO [41] is adopted to solve the NMPC motion planning.

VI. Simulation

A. Simulation Setting

In this work, driving style is considered in decision making, thus, we should firstly validate the proposed driving style estimation algorithm. Then human driver should participate in the experiment to validate the driving style estimation algorithm and decision making approach. In this section, a three-vehicle simulation and a two-vehicle HIL simulation is designed to verify the performance of proposed algorithm. HIL simulation is a commonly used method that validates the research concerning human driver. As is shown in Fig. 5 the HIL simulation environment includes a CARLA simulator and a Logitech G29 steering wheel with a pedal. The detail of CARLA simulator can be found in [41]. In our simulation environment, human participant can drive a virtual vehicle

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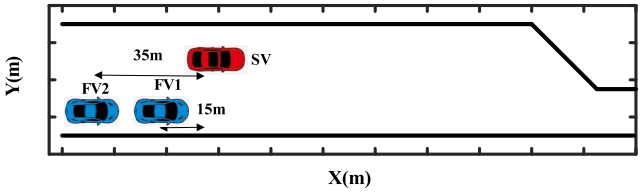


Fig. 6. Merging scenario for three-vehicle simulation.

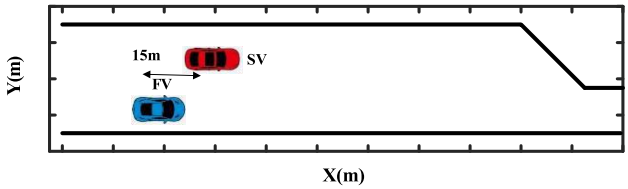


Fig. 7. Merging scenario for two-vehicle HIL simulation.

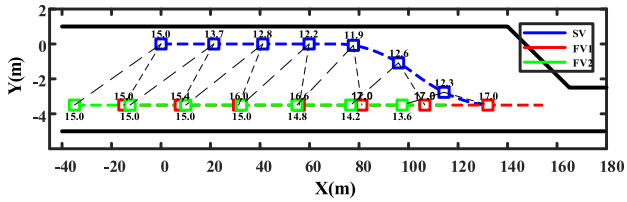


Fig. 8. Trajectory result of FM ($\rho_1 = 0.4$, $\rho_2 = 0.7$).

containing real engine (or motor), transmission and tire model. It is worth noting that CARLA simulator only provide python interface. Thus we build a shared library of motion planning code written in C++, and call the functions via python library ctypes [42]. In addition, a low level controller is adopted to track the target velocity and steering angle. We also compare our decision model with decision making model ignoring interaction or human driver's driving style to validate the effectiveness of our approach.

B. Simulation

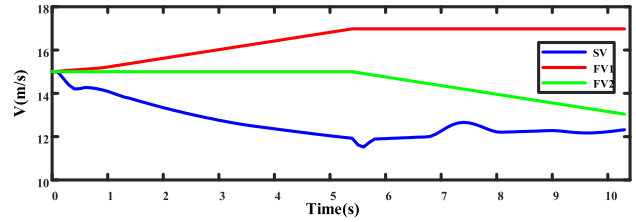
The three-vehicle simulation scenario is designed as Fig. 6. In this simulation, the driving style of FV1 and FV2 can be given arbitrarily. By comparing the estimated driving styles of the two following vehicles and the given driving style. In addition, we compare our full model (FM) with two decision model: (1) FM without consideration of the interaction with FV (NIM); (2) FM without consideration of driving style (NDM). In terms of NDM, SV assumes that FV's driving style is normal.

By comparing the three model, the effectiveness of the proposed MBC system can be proved.

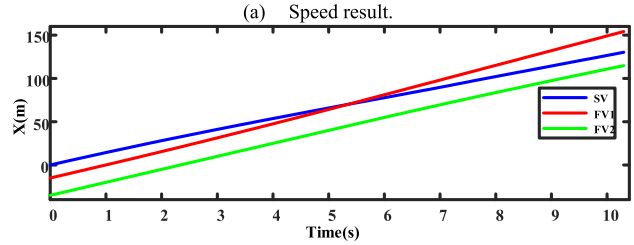
The human-in-the-loop simulation scenario is designed as Fig. 7. In the initial step, FV runs at a constant speed and SV turn on its turning signal, then both participants make decision and control vehicle. If FV overtakes SV, SV will choose a virtual vehicle following far behind. The HIL simulation can reveal how the driving style of a human driver changes during the whole merging process.

C. The Three-Vehicle Simulation Result

The driving style of FV1 and FV2 are set to 0.4 and 0.7 respectively at first. As is depicted in Fig.10(b), the estimated

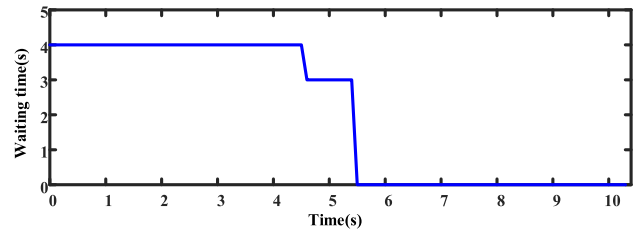


(a) Speed result.

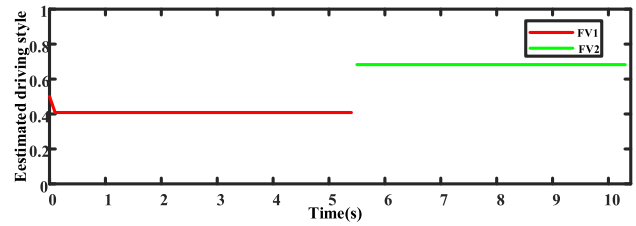


(b) Longitudinal displacement result.

Fig. 9. Trajectory result of FM ($\rho_1 = 0.4$, $\rho_2 = 0.7$).



(a) Waiting time result.



(b) Estimated driving style

Fig. 10. Behavior control and driving style estimation result of FM ($\rho_1 = 0.4$, $\rho_2 = 0.7$).

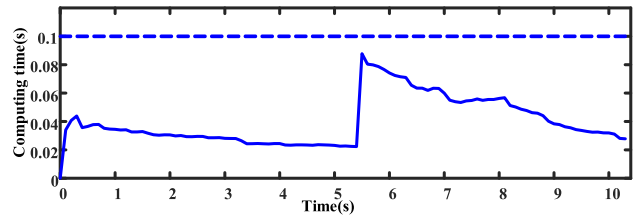


Fig. 11. Computing time of merging behavior control system ($\rho_1 = 0.4$, $\rho_2 = 0.7$).

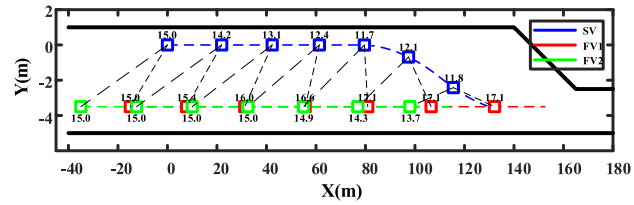


Fig. 12. Trajectory result of NIM ($\rho_1 = 0.4$, $\rho_2 = 0.7$).

driving style of FV1 is quickly measured to be 0.409, which is very close to the real driving style of FV1. It can be seen that SV first decelerates to yield and prepare to merge between FV1 and FV2 in Fig. 9(a). After FV1 overtaking SV,

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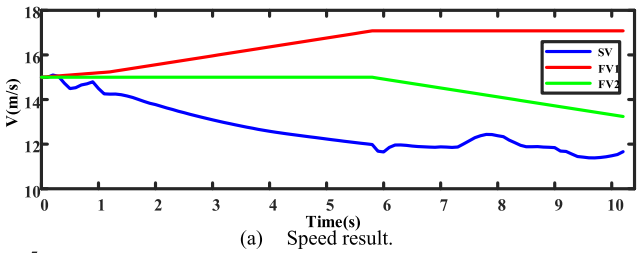
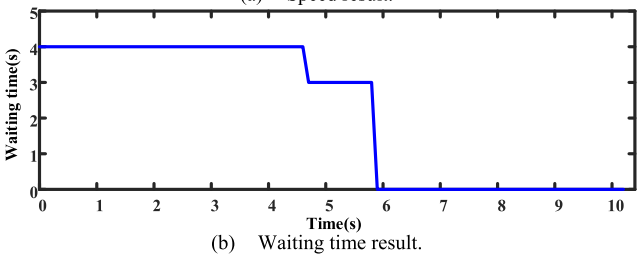


(a) Speed result.



(b) Waiting time result.

Fig. 13. Trajectory result of NIM ($\rho_1 = 0.4$, $\rho_2 = 0.7$).

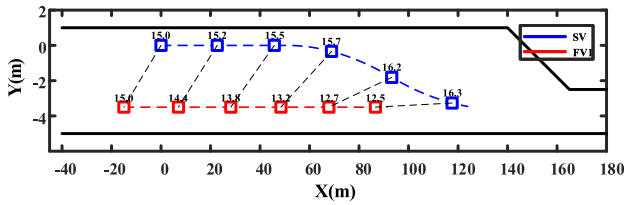
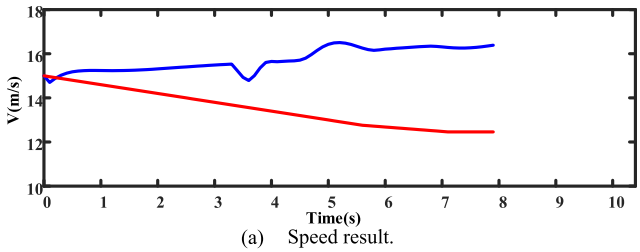
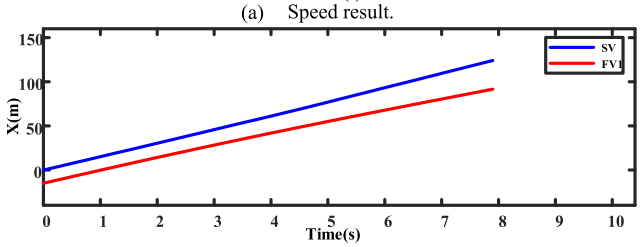


Fig. 14. Trajectory result of FM ($\rho_1 = 0.7$, $\rho_2 = 0.7$).



(a) Speed result.

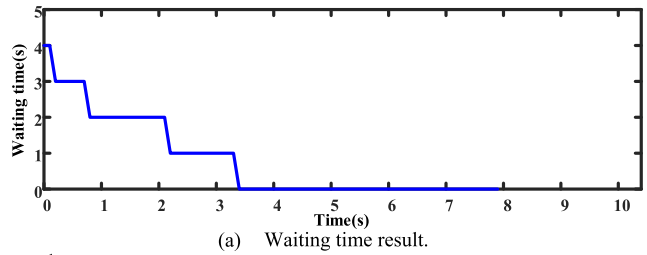


(b) Longitudinal displacement result.

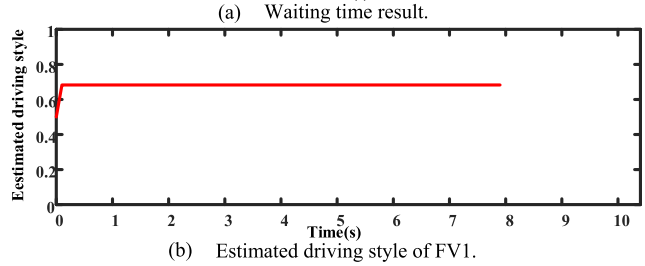
Fig. 15. Trajectory result of FM ($\rho_1 = 0.7$, $\rho_2 = 0.7$).

SV starts to interact with FV2, and the estimated driving style of FV2 is 0.683. By interacting with FV2, SV chooses to merge immediately. Due to the lateral acceleration constraint, SV decelerates slightly, and later SV accelerates to keep the distance between SV and FV2 until the merging process is finished. During this process, FV2 chooses to decelerate out of the safety preference. In the whole merging process, the computing time is lower than 0.1(i.e. time threshold), which is shown in Fig. 17. Therefore, the proposed MBC system can be run in real time.

The simulation result of NIM is similar to FM in this case, but due to the lack of consideration of interaction, SV's speed fluctuates at the beginning, which can be seen in Fig. 13(a). As is shown in Fig. 13(b), after FV1 overtaking



(a) Waiting time result.



(b) Estimated driving style of FV1.

Fig. 16. Behavior control and driving style estimation result of FM ($\rho_1 = 0.7$, $\rho_2 = 0.7$).

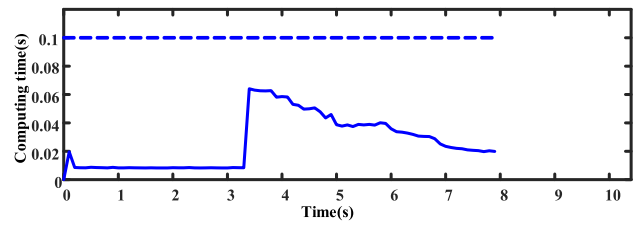


Fig. 17. Computing time of merging behavior control system ($\rho_1 = 0.7$, $\rho_2 = 0.7$).

SV, SV chooses FV2 as the game participant. The rest of merging process is similar to FM. And the merging time of SV with NIM is slightly later than the time of SV with FM. By comparing FM and NIM, we can draw the conclusion that considering interaction indeed proves the decision quality. The NDM model performs similarly to NIM, we won't go into details here.

But in another case that the real driving style of FV1 is set to 0.7, NDM has poor performance. In this case, FM still can estimate the driving style of FV1 accurately. As FV1 decelerates, SV chooses to accelerate slightly to gain a safer distance to merge. At the beginning of merging maneuver, SV decelerates slightly to keep a low lateral acceleration, then SV accelerates to maintain a safe distance from FV1 until the finishing the merging maneuver. As is depicted in Fig. 17, the computing time of each step is lower than the time threshold.

In this case, SV with NIM can not estimate FV1's driving style, thus SV tries to decelerate to yield to FV1. Meanwhile, FV1 also decelerates to yield to SV. Both SV and FV1 decelerates to yield to each other, so FV can not maintain a safe gap, thus fail to merge. By comparing FM and NDM in this case, we can see the significance of accurately estimating driving style of FV.

D. The Two-Vehicle HIL Simulation Result

The HIL simulation reveals that driver's driving style varies during the merging process. As is shown in Fig. 20, driving style is not a constant in the HIL simulation. As a matter of fact, driver's preference to safety and efficiency would change as the driving scenario changes. It can be seen from

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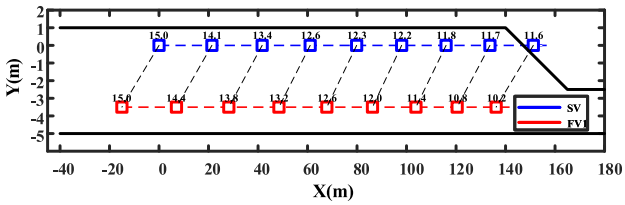

Fig. 18. Trajectory result of NDM ($\rho_1 = 0.7$, $\rho_2 = 0.7$).

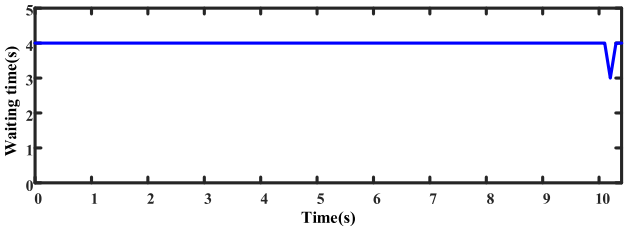

Fig. 19. Waiting time result of NDM ($\rho_1 = 0.7$, $\rho_2 = 0.7$).

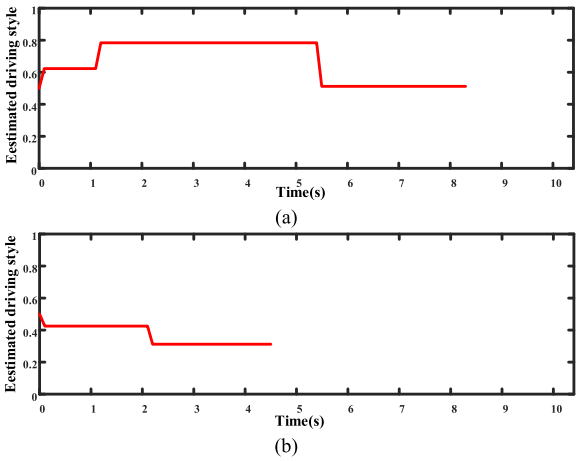

Fig. 20. Estimated driving style of different participants.

Fig.20(a), the driver in FV first choose to drive carefully, thus the estimated driving style is larger than 0.5. After the driver in FV noticing that SV accelerates, FV chooses a more cautious decision, so the estimated driving style increases. In another situation, FV chooses to overtake SV as FV prefers higher travel efficiency. When the FV is approaching SV, FV may adopt a higher acceleration, which means SV prefer higher travel efficiency. The above process can be found in Fig 20(b). The driving style estimation result indicates that the assumption of fixed driving style is unrealistic and online estimation should be adopted.

VII. Conclusion

In this paper, a merging behavior control system considering social interaction is presented. The decision model utilizes Stackelberg game to consider the social interaction between vehicles. In terms of the driver's heterogeneity, an online driving style estimation algorithm is developed, which plays a significant role in predicting following vehicle's decision. Then decision making module is combined with motion planning module, which outputs the optimal trajectories in real time. In order to verify the proposed merging behavior control system with driving style estimation, a three-vehicle simulation and a two-vehicle HIL simulation are designed. The test result shows that: 1) Our system is able to consider interaction between ego vehicle and surrounding vehicles. 2) The proposed driving style estimation algorithm can estimate driving style of following vehicle accurately, which can prove the decision making with the consideration of interaction. 3) The proposed MBC system can be run in real time. In addition, the HIL simulation shows that driver's driving style might change in merging scenario, which indicates the importance of online driving style estimation.

In merging maneuver, autonomous vehicle only has one target lane to make lane changing decision. In the future wok, we will extend the proposed decision making approach to the general mandatory lane changing with lane selection. In this work, we utilize a two-player non-cooperative game. In the future work, we will consider multiplayer game in mandatory lane changing scenario.

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